

DIOGO GOMES DE ARAÚJO

(351) 914 576 371 | contact@diogoaraujo.com | diogoaraujo.com | linkedin.com/in/diogogomesaraujo | github.com/diogogomesaraujo

EDUCATION

Master's

Computer Science

Sep 2025 – Present

Universidade do Porto (FCUP)

Bachelor's

Computer Science and Engineering

Sep 2022 – Jul 2025

Universidade da Beira Interior

WORK EXPERIENCE

INESC TEC

Researcher

Jan 2026 – Apr 2026

Hybrid

- Developed and maintained **IoT gateway microservices** for Distributed Energy Resources communication.
- Deployed a **Zero Trust framework** (Teleport) for **secure remote access to all IoT devices** in the Power and Energy Systems laboratory.

STAR Junior Enterprise

Project Manager and Backend Developer

May 2024 – Aug 2025

Remote

- Developed and maintained **backend REST API's** for mobile and web applications.
- Assisted in recruiting by **screening candidates, directed teams and worked with clients** to deliver projects on time and according to the requirements.

PROJECTS

Group Wallet for Nano | Rust, Tokio, Dioxus, Dalek

- Developed an **open-source library** with a custom implementation of FROST, **enabling shared account creation and group transactions** on Nano's blockchain.
- Leveraged modern cryptographic protocols to **minimize communication rounds required for transaction signing** on Nano's network.
- Created a **desktop application** that uses the custom library to allow groups to perform transactions remotely in an **intuitive and secure graphical environment**.

PoW Blockchain from Decentralized Auctions | Rust, Tokio, LibP2P, Dalek, Docker

- Developed a DHT overlay based on Kademlia for **decentralized network construction**.
- PoW Blockchain with **Blake2b hashing, Ed25519 signatures, and Merkle Trees** for transaction validation.
- Client for auctions** using the blockchain layer as a public ledger.
- Docker **autonomous network behaviour simulation** with clients, bootstrap nodes, well behaved nodes, and malicious nodes.

CERTIFICATIONS

2nd Place in IEEE RetroJam 2025 | IEEE UP Student Branch

Oct 2025

- Built a game in 48 hours from scratch, using Rust and Raylib, learning how to **reduce a problem to its core requirements**.

Participation in Inforum 2025 | Inforum

Sep 2025

- Presented my Bachelor's Degree final project to an **audience of researchers, professionals, and students**, receiving valuable feedback and constructive criticism on my work.

Industrial Robotics Fundamentals | Universidade da Beira Interior

Jul 2025

Volunteering | Missão País

Feb 2024

Participation in RoboCup 2016 | RoboCup Federation

Jun 2016

1st Place in the National Robotics Championship | Instituto Politécnico de Bragança

May 2016

TECHNICAL SKILLS

Programming Languages: Rust, OCaml, C, Haskell, Java, SQL

Frameworks & Tools: Git, UNIX, Tokio, POSIX, STM, OpenMP, gRPC, QuickCheck, QUIC, LibP2P, Dalek, Raylib, PostgreSQL